

NAME

ExtendedConnectivityFingerprints.pl - Generate extended connectivity fingerprints for SD files

SYNOPSIS

ExtendedConnectivityFingerprints.pl SDFFile(s)...

```
ExtendedConnectivityFingerprints.pl [--AromaticityModel AromaticityModelType] [-a, --AtomIdentifierType
AtomicInvariantsAtomTypes] [--AtomicInvariantsToUse "AtomicInvariant,AtomicInvariant..."] [
--FunctionalClassesToUse "FunctionalClass1,FunctionalClass2..."] [--BitsOrder Ascending | Descending] [-b,
--BitStringFormat BinaryString | HexadecimalString] [--CompoundID DataFieldName or LabelPrefixString] [
--CompoundIDLabel text] [--CompoundIDMode] [--DataFields "FieldLabel1,FieldLabel2,..."] [-d,
--DataFieldsMode All | Common | Specify | CompoundID] [-f, --Filter Yes | No] [--FingerprintsLabel text] [-h,
--help] [-k, --KeepLargestComponent Yes | No] [-m, --mode ExtendedConnectivity | ExtendedConnectivityCount |
ExtendedConnectivityBits] [-n, --NeighborhoodRadius number] [--OutDelim comma | tab | semicolon] [--output
SD | FP | text | all] [-o, --overwrite] [-q, --quote Yes | No] [-r, --root RootName] [-s, --size number] [
--UsePerlCoreRandom Yes | No] [-v, --VectorStringFormat IDsAndValuesString | IDsAndValuesPairsString |
ValuesAndIDsString | ValuesAndIDsPairsString] [-w, --WorkingDir dirname] SDFFile(s)...
```

DESCRIPTION

Generate extended connectivity fingerprints [Ref 48, Ref 52] for *SDFFile(s)* and create appropriate SD, FP or CSV/TSV text file(s) containing fingerprints vector strings corresponding to molecular fingerprints.

Multiple *SDFFile* names are separated by spaces. The valid file extensions are *.sdf* and *.sd*. All other file names are ignored. All the SD files in a current directory can be specified either by **.sdf* or the current directory name.

The current release of MayaChemTools supports generation of extended connectivity fingerprints corresponding to following *-a, --AtomIdentifierTypes*:

```
AtomicInvariantsAtomTypes, DREIDINGAtomTypes, EStateAtomTypes,
FunctionalClassAtomTypes, MMFF94AtomTypes, SLogPAtomTypes,
SYBYLAtomTypes, TPSAAtomTypes, UFFAtomTypes
```

Based on values specified for *-a, --AtomIdentifierType*, *--AtomicInvariantsToUse* and *--FunctionalClassesToUse*, initial atom types are assigned to all non-hydrogen atoms in a molecule and these atom types strings are converted into initial atom identifier integers using `TextUtil::HashCode` function. The duplicate atom identifiers are removed.

For *-n, --NeighborhoodRadius* value of 0, the initial set of unique atom identifiers comprises the molecule fingerprints. Otherwise, atom neighborhoods are generated for each non-hydrogen atom up to specified *-n, --NeighborhoodRadius* value. For each non-hydrogen central atom at a specific radius, its neighbors at next radius level along with their bond orders and previously calculated atom identifiers are collected which in turn are used to generate a new integer atom identifier; the bond orders and atom identifier pairs list is first sorted by bond order followed by atom identifiers to make these values graph invariant.

After integer atom identifiers have been generated for all non-hydrogen atoms at all specified neighborhood radii, the duplicate integer atom identifiers corresponding to same hash code value generated using `TextUtil::HashCode` are tracked by keeping the atom identifiers at lower radius. Additionally, all structurally duplicate integer atom identifiers at each specified radius are also tracked by identifying equivalent atoms and bonds corresponding to substructures used for generating atom identifier and keeping integer atom identifier with lowest value.

For *ExtendedConnectivity* value of fingerprints *-m, --mode*, the duplicate identifiers are removed from the list and the unique atom identifiers constitute the extended connectivity fingerprints of a molecule.

For *ExtendedConnectivityCount* value of fingerprints *-m, --mode*, the occurrence of each unique atom identifiers appears is counted and the unique atom identifiers along with their count constitute the extended connectivity fingerprints of a molecule.

For *ExtendedConnectivityBits* value of fingerprints *-m, --mode*, the unique atom identifiers are used as a random number seed to generate a random integer value between 0 and *--Size* which in turn is used to set corresponding bits in the fingerprint bit-vector string.

Example of *SD* file containing extended connectivity fingerprints string data:

```
... ..
... ..
$$$$
... ..
... ..
```

```

... ..
41 44 0 0 0 0 0 0 0 0 0999 V2000
-3.3652 1.4499 0.0000 C 0 0 0 0 0 0 0 0 0 0
... ..
2 3 1 0 0 0 0
... ..
M END
> <CmpdID>
Cmpd1

> <ExtendedConnectivityFingerprints>
FingerprintsVector;ExtendedConnectivity:AtomicInvariantsAtomTypes:Radiu
s2;60;AlphaNumericalValues;ValuesString;73555770 333564680 352413391 66
6191900 1001270906 1371674323 1481469939 1977749791 2006158649 21414087
99 49532520 64643108 79385615 96062769 273726379 564565671 855141035 90
6706094 988546669 1018231313 1032696425 1197507444 1331250018 133853...

$$$$
... ..
... ..

```

Example of *FP* file containing extended connectivity fingerprints string data:

```
#
# FingerprintsStringType = FingerprintsVector
#
# Description = ExtendedConnectivity:AtomicInvariantsAtomTypes:Radius2
# VectorStringFormat = ValuesString
# VectorValuesType = AlphaNumericalValues
#
Cmpd1 60;73555770 333564680 352413391 666191900 1001270906 137167432...
Cmpd2 41;73555770 333564680 666191900 1142173602 1363635752 14814699...
... ..
... ..
```

Example of CSV *Text* file containing extended connectivity fingerprints string data:

```
"CompoundID", "ExtendedConnectivityFingerprints"
"Cmpd1", "FingerprintsVector;ExtendedConnectivity:AtomicInvariantsAtomTy
pes:Radius2;60;AlphaNumericalValues;ValuesString;73555770 333564680 352
413391 666191900 1001270906 1371674323 1481469939 1977749791 2006158649
2141408799 49532520 64643108 79385615 96062769 273726379 564565671 8551
41035 906706094 988546669 1018231313 1032696425 1197507444 13312500..."
... ..
... ..
```

The current release of MayaChemTools generates the following types of extended connectivity fingerprints vector strings:

```
FingerprintsVector;ExtendedConnectivity:AtomicInvariantsAtomTypes:Radius2;60;AlphaNumericalValues;ValuesString;73555770 333564680 352413391
666191900 1001270906 1371674323 1481469939 1977749791 2006158649 21414
08799 49532520 64643108 79385615 96062769 273726379 564565671 85514103
5 906706094 988546669 1018231313 1032696425 1197507444 1331250018 1338
532734 1455473691 1607485225 1609687129 1631614296 1670251330 17303...
```

```
FingerprintsVector;ExtendedConnectivityCount:AtomicInvariantsAtomTypes
:Radius2;60;NumericalValues;IDsAndValuesString;73555770 333564680 3524
13391 666191900 1001270906 1371674323 1481469939 1977749791 2006158649
2141408799 49532520 64643108 79385615 96062769 273726379 564565671...;
3 2 1 1 14 1 2 10 4 3 1 1 1 2 2 1 1 2 3 1 2 1 3 3 8 2 2 2 6 2
1 2 1 1 2 1 1 2 1 2 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

[illegible]

1668649038 1684198062 1812312554 1859266290 1891651106 2072549404 ...

FingerprintsVector;ExtendedConnectivity:TPSAAtomTypes:Radius2;47;Alpha
NumericalValues;ValuesString;20818206 259344053 862102353 1331904542 1
700688206 265614156 363161397 681332588 810600886 885767127 950172500
951454814 1059668746 1247054493 1382302230 1399502637 1805025917 19189
39561 2114677228 2126402271 8130483 17645742 32278373 149975755 160327
654 256360355 279492740 291251259 317592700 333763396 972105960 101...

FingerprintsVector;ExtendedConnectivity:UFFAtomTypes:Radius2;56;Alpha
NumericalValues;ValuesString;280305427 357928343 721790579 1151822898 1
207111054 1380963747 1568213839 1603445250 4559268 55012922 180940813
335715751 534801009 684609658 829361048 972945982 999881534 1007655741
1213692591 1222032501 1224517934 1235687794 1244268533 1528120700 162
9595024 1856308891 1978806036 2001865095 2096549435 172675415 18344...

OPTIONS

--AromaticityModel *MDLAromaticityModel* | *TriposAromaticityModel* | *MMFFAromaticityModel* |
ChemAxonBasicAromaticityModel | *ChemAxonGeneralAromaticityModel* | *DaylightAromaticityModel* |
MayaChemToolsAromaticityModel

Specify aromaticity model to use during detection of aromaticity. Possible values in the current release are: *MDLAromaticityModel*, *TriposAromaticityModel*, *MMFFAromaticityModel*, *ChemAxonBasicAromaticityModel*, *ChemAxonGeneralAromaticityModel*, *DaylightAromaticityModel* or *MayaChemToolsAromaticityModel*. Default value: *MayaChemToolsAromaticityModel*.

The supported aromaticity model names along with model specific control parameters are defined in *AromaticityModelsData.csv*, which is distributed with the current release and is available under *lib/data* directory. *Molecule.pm* module retrieves data from this file during class instantiation and makes it available to method *DetectAromaticity* for detecting aromaticity corresponding to a specific model.

-a, --AtomIdentifierType *AtomicInvariantsAtomTypes* | *FunctionalClassAtomTypes* | *DREIDINGAtomTypes* |
EStateAtomTypes | *MMFF94AtomTypes* | *SLogPAtomTypes* | *SYBYLAtomTypes* | *TPSAAtomTypes* | *UFFAtomTypes*

Specify atom identifier type to use for assignment of initial atom identifier to non-hydrogen atoms during calculation of extended connectivity fingerprints [Ref 48, Ref 52]. Possible values in the current release are: *AtomicInvariantsAtomTypes*, *FunctionalClassAtomTypes*, *DREIDINGAtomTypes*, *EStateAtomTypes*, *MMFF94AtomTypes*, *SLogPAtomTypes*, *SYBYLAtomTypes*, *TPSAAtomTypes*, *UFFAtomTypes*. Default value: *AtomicInvariantsAtomTypes*.

--AtomicInvariantsToUse "AtomicInvariant,AtomicInvariant..."

This value is used during *AtomicInvariantsAtomTypes* value of a, --AtomIdentifierType option. It's a list of comma separated valid atomic invariant atom types.

Possible values for atomic invariants are: AS, X, BO, LBO, SB, DB, TB, H, Ar, RA, FC, MN, SM. Default value [Ref 24]: AS,X,BO,H,FC,MN.

The atomic invariants abbreviations correspond to:

AS = Atom symbol corresponding to element symbol

X<n> = Number of non-hydrogen atom neighbors or heavy atoms

BO<n> = Sum of bond orders to non-hydrogen atom neighbors or heavy atoms

LBO<n> = Largest bond order of non-hydrogen atom neighbors or heavy atoms

SB<n> = Number of single bonds to non-hydrogen atom neighbors or heavy atoms

DB<n> = Number of double bonds to non-hydrogen atom neighbors or heavy atoms

TB<n> = Number of triple bonds to non-hydrogen atom neighbors or heavy atoms

H<n> = Number of implicit and explicit hydrogens for atom

Ar = Aromatic annotation indicating whether atom is aromatic

RA = Ring atom annotation indicating whether atom is a ring

FC<+n/-n> = Formal charge assigned to atom

MN<n> = Mass number indicating isotope other than most abundant isotope

SM<n> = Spin multiplicity of atom. Possible values: 1 (singlet), 2 (doublet) or 3 (triplet)

Atom type generated by *AtomTypes::AtomicInvariantsAtomTypes* class corresponds to:

AS.X<n>.BO<n>.LBO<n>.<SB><n>.<DB><n>.<TB><n>.H<n>.Ar.RA.FC<+n/-n>.MN<n>.SM<n>

Except for AS which is a required atomic invariant in atom types, all other atomic invariants are

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```
X : NumOfNonHydrogenAtomNeighbors or NumOfHeavyAtomNeighbors
BO : SumOfBondOrdersToNonHydrogenAtoms or SumOfBondOrdersToHeavyAtoms
LBO : LargestBondOrderToNonHydrogenAtoms or LargestBondOrderToHeavyAtoms
SB : NumOfSingleBondsToNonHydrogenAtoms or NumOfSingleBondsToHeavyAtoms
DB : NumOfDoubleBondsToNonHydrogenAtoms or NumOfDoubleBondsToHeavyAtoms
TB : NumOfTripleBondsToNonHydrogenAtoms or NumOfTripleBondsToHeavyAtoms
H : NumOfImplicitAndExplicitHydrogens
Ar : Aromatic
RA : RingAtom
FC : FormalCharge
MN : MassNumber
SM : SpinMultiplicity
```

--BitsOrder *Ascending | Descending*

Ascending bit order which corresponds to first bit in each byte as the lowest bit as opposed to the highest bit.

-b, --BitStringFormat *BinaryString* | *HexadecimalString*

BinaryString corresponds to an ASCII string containing 1s and 0s. *HexadecimalString* contains bit values in ASCII hexadecimal format.

[illegible][illegible]

This value is used during *FunctionalClassAtomTypes* value of a, --AtomIdentifierType option. It's a list of comma separated valid functional classes.

Possible values for atom functional classes are: *Ar, CA, H, HBA, HBD, Hal, NI, PI, RA*. Default value [Ref 24]: *HBD,HBA,PI,NI,Ar,Hal*.

```
HBD: HydrogenBondDonor
HBA: HydrogenBondAcceptor
PI : PositivelyIonizable
NI : NegativelyIonizable
Ar : Aromatic
Hal : Halogen
H : Hydrophobic
RA : RingAtom
CA : ChainAtom
```

Functional class atom type specification for an atom corresponds to:

Ar.CA.H.HBA.HBD.Hal.NI.PI.RA

AtomTypes::FunctionalClassAtomTypes module is used to assign functional class atom types. It uses following definitions [Ref 60-61, Ref 65-66]:

HydrogenBondDonor: NH, NH₂, OH
 HydrogenBondAcceptor: N[!H], O
 PositivelyIonizable: +, NH₂
 NegativelyIonizable: -, C(=O)OH, S(=O)OH, P(=O)OH

--CompoundID *DataFieldName* or *LabelPrefixString*

This value is --CompoundIDMode specific and indicates how compound ID is generated.

For *DataField* value of --CompoundIDMode option, it corresponds to datafield label name whose value is used as compound ID; otherwise, it's a prefix string used for generating compound IDs like *LabelPrefixString*<Number>. Default value, *Cmpd*, generates compound IDs which look like *Cmpd*<Number>.

Examples for *DataField* value of --CompoundIDMode:

MolID
 ExtReg

Examples for *LabelPrefix* or *MolNameOrLabelPrefix* value of --CompoundIDMode:

Compound

The value specified above generates compound IDs which correspond to *Compound*<Number> instead of default value of *Cmpd*<Number>.

--CompoundIDLabel text

Specify compound ID column label for FP or CSV/TSV text file(s) used during *CompoundID* value of --DataFieldsMode option. Default: *CompoundID*.

--CompoundIDMode *DataField* | *MolName* | *LabelPrefix* | *MolNameOrLabelPrefix*

Specify how to generate compound IDs and write to FP or CSV/TSV text file(s) along with generated fingerprints for *FP* | *text* | *all* values of --output option: use a *SDF* file(s) datafield value; use molname line from *SDF* file(s); generate a sequential ID with specific prefix; use combination of both *MolName* and *LabelPrefix* with usage of *LabelPrefix* values for empty molname lines.

Possible values: *DataField* | *MolName* | *LabelPrefix* | *MolNameOrLabelPrefix*. Default: *LabelPrefix*.

For *MolNameAndLabelPrefix* value of --CompoundIDMode, molname line in *SDF* file(s) takes precedence over sequential compound IDs generated using *LabelPrefix* and only empty molname values are replaced with sequential compound IDs.

This is only used for *CompoundID* value of --DataFieldsMode option.

--DataFields "*FieldLabel1,FieldLabel2,...*"

Comma delimited list of *SDF* file(s) data fields to extract and write to CSV/TSV text file(s) along with generated fingerprints for *text* | *all* values of --output option.

This is only used for *Specify* value of --DataFieldsMode option.

Examples:

Extreg
 MolID, CompoundName

-d, --DataFieldsMode *All* | *Common* | *Specify* | *CompoundID*

Specify how data fields in *SDF* file(s) are transferred to output CSV/TSV text file(s) along with generated fingerprints for *text* | *all* values of --output option: transfer all SD data field; transfer SD data files common to all compounds; extract specified data fields; generate a compound ID using molname line, a compound prefix, or a combination of both. Possible values: *All* | *Common* | *Specify* | *CompoundID*. Default value: *CompoundID*.

-f, --Filter Yes | No

Specify whether to check and filter compound data in *SDF* file(s). Possible values: *Yes* or *No*. Default value: *Yes*.

By default, compound data is checked before calculating fingerprints and compounds containing atom

data corresponding to non-element symbols or no atom data are ignored.

--FingerprintsLabel *text*

SD data label or text file column label to use for fingerprints string in output SD or CSV/TSV text file(s) specified by --output. Default value: *ExtendedConnectivityFingerprints*.

-h, --help

Print this help message.

-k, --KeepLargestComponent *Yes | No*

Generate fingerprints for only the largest component in molecule. Possible values: *Yes* or *No*. Default value: *Yes*.

For molecules containing multiple connected components, fingerprints can be generated in two different ways: use all connected components or just the largest connected component. By default, all atoms except for the largest connected component are deleted before generation of fingerprints.

-m, --mode *ExtendedConnectivity | ExtendedConnectivityCount | ExtendedConnectivityBits*

Specify type of extended connectivity fingerprints to generate for molecules in *SDF* file(s). Possible values: *ExtendedConnectivity*, *ExtendedConnectivityCount* or *ExtendedConnectivityBits*. Default value: *ExtendedConnectivity*.

For *ExtendedConnectivity* value of fingerprints -m, --mode, a fingerprint vector containing unique atom identifiers constitute the extended connectivity fingerprints of a molecule.

For *ExtendedConnectivityCount* value of fingerprints -m, --mode, a fingerprint vector containing unique atom identifiers along with their count constitute the extended connectivity fingerprints of a molecule.

For *ExtendedConnectivityBits* value of fingerprints -m, --mode, a fingerprint bit vector indicating presence/absence of structurally unique atom identifiers constitute the extended connectivity fingerprints of a molecule.

-n, --NeighborhoodRadius *number*

Atomic neighborhood radius for generating extended connectivity neighborhoods. Default value: 2. Valid values: ≥ 0 . Neighborhood radius of zero correspond to just the list of non-hydrogen atoms.

Default value of 2 for atomic neighborhood radius generates extended connectivity fingerprints corresponding to path length or diameter value of 4 [Ref 52b].

--OutDelim *comma | tab | semicolon*

Delimiter for output CSV/TSV text file(s). Possible values: *comma*, *tab*, or *semicolon*. Default value: *comma*.

--output *SD | FP | text | all*

Type of output files to generate. Possible values: *SD*, *FP*, *text*, or *all*. Default value: *text*.

-o, --overwrite

Overwrite existing files.

-q, --quote *Yes | No*

Put quote around column values in output CSV/TSV text file(s). Possible values: *Yes* or *No*. Default value: *Yes*.

-r, --root *RootName*

New file name is generated using the root: *<Root>.<Ext>*. Default for new file names: *<SDFFileName><ExtendedConnectivityFP>.<Ext>*. The file type determines *<Ext>* value. The *sdf*, *fpf*, *csv*, and *tsv* *<Ext>* values are used for SD, FP, comma/semicolon, and tab delimited text files, respectively. This option is ignored for multiple input files.

-s, --size *number*

Size of bit-vector to use during generation of fingerprints bit-vector string for *ExtendedConnectivityBits* value of -m, --mode. Default value: 1024. Valid values correspond to any positive integer which satisfies the following criteria: power of 2, ≥ 32 and $\leq 2^{32}$.

Examples:

512
1024
2048

--UsePerlCoreRandom *Yes | No*

Specify whether to use Perl CORE::rand or MayaChemTools MathUtil::random function during random number generation for setting bits in fingerprints bit-vector strings. Possible values: *Yes* or *No*. Default value: *Yes*.

No value option for --UsePerlCoreRandom allows the generation of fingerprints bit-vector strings which are same across different platforms.

The random number generator implemented in MayaChemTools is a variant of linear congruential generator (LCG) as described by Miller et al. [Ref 120]. It is also referred to as Lehmer random number generator or Park-Miller random number generator.

Unlike Perl's core random number generator function rand, the random number generator implemented in MayaChemTools, MathUtil::random, generates consistent random values across different platforms for a specific random seed and leads to generation of portable fingerprints bit-vector strings.

-v, --VectorStringFormat *ValuesString* | *IDsAndValuesString* | *IDsAndValuesPairsString* | *ValuesAndIDsString* | *ValuesAndIDsPairsString*

Format of fingerprints vector string data in output SD, FP or CSV/TSV text file(s) specified by --output used during <ExtendedConnectivityCount> value of -m, --mode option. Possible values: *ValuesString*, *IDsAndValuesString* | *IDsAndValuesPairsString* | *ValuesAndIDsString* | *ValuesAndIDsPairsString*.

Default value during <ExtendedConnectivityCount> value of -m, --mode option: *IDsAndValuesString*.

Default value during <ExtendedConnectivity> value of -m, --mode option: *ValuesString*.

Examples:

```
FingerprintsVector;ExtendedConnectivity:AtomicInvariantsAtomTypes:Radius2;60;AlphaNumericalValues;ValuesString;73555770 333564680 352413391
666191900 1001270906 1371674323 1481469939 1977749791 2006158649 21414
08799 49532520 64643108 79385615 96062769 273726379 564565671 85514103
5 906706094 988546669 1018231313 1032696425 1197507444 1331250018 1338
532734 1455473691 1607485225 1609687129 1631614296 1670251330 17303...
```

```
FingerprintsVector;ExtendedConnectivityCount:AtomicInvariantsAtomTypes
:Radius2;60;NumericalValues;IDsAndValuesString;73555770 333564680 3524
13391 666191900 1001270906 1371674323 1481469939 1977749791 2006158649
2141408799 49532520 64643108 79385615 96062769 273726379 564565671...;
3 2 1 1 14 1 2 10 4 3 1 1 1 2 1 2 1 1 1 2 3 1 1 2 1 3 3 8 2 2 2 6 2
1 2 1 1 2 1 1 1 2 1 1 2 1 1 1 1 1 1 1 1 1 2 1 1
```

-w, --WorkingDir *DirName*

Location of working directory. Default: current directory.

EXAMPLES

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using atomic invariants atom types in vector string format and create a SampleECAIFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -r SampleECAIFP -o Sample.sdf
```

To generate extended connectivity count fingerprints corresponding to neighborhood radius up to 2 using atomic invariants atom types in vector string format and create a SampleECAIFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -m ExtendedConnectivityCount
-r SampleECAIFP -o Sample.sdf
```

To generate extended connectivity bits fingerprints as hexadecimal bit-string corresponding to neighborhood radius up to 2 using atomic invariants atom types in vector string format and create a SampleECAIFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -m ExtendedConnectivityBits
-r SampleECAIFP -o Sample.sdf
```

To generate extended connectivity bits fingerprints as binary bit-string corresponding to neighborhood radius up to 2 using atomic invariants atom types in vector string format and create a SampleECAIFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -m ExtendedConnectivityBits
--BitStringFormat BinaryString -r SampleECAIFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using atomic invariants atom types in vector string format and create SampleECAIFP.sdf, SampleECAIFP.fpf and SampleECAIFP.csv files containing sequential compound IDs in CSV file along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl --output all -r SampleECAIFP  
-o Sample.sdf
```

To generate extended connectivity count fingerprints corresponding to neighborhood radius up to 2 using atomic invariants atom types in vector string format and create SampleECAIFP.sdf, SampleECAIFP.fpf and SampleECAIFP.csv files containing sequential compound IDs in CSV file along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -m ExtendedConnectivityCount  
--output all -r SampleECAIFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using functional class atom types in vector string format and create a SampleECFCFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a FunctionalClassAtomTypes  
-r SampleECFCFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using DREIDING atom types in vector string format and create a SampleECFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a DREIDINGAtomTypes  
-r SampleECFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using E-state atom types in vector string format and create a SampleECFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a EStateAtomTypes  
-r SampleECFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using MMFF94 atom types in vector string format and create a SampleECFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a MMFF94AtomTypes  
-r SampleECFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using SLogP atom types in vector string format and create a SampleECFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a SLogPAtomTypes  
-r SampleECFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using SYBYL atom types in vector string format and create a SampleECFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a SYBYLAtomTypes  
-r SampleECFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using TPSA atom types in vector string format and create a SampleECFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a TPSAAtomTypes  
-r SampleECFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using UFF atom types in vector string format and create a SampleECFP.csv file containing sequential compound IDs along with

fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a UFFAtomTypes  
-r SampleECFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 3 using atomic invariants atom types in vector string format and create a SampleECAIFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a AtomicInvariantsAtomTypes -n 3  
-r SampleECAIFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 3 using functional class atom types in vector string format and create a SampleECFCFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a FunctionalClassAtomTypes -n 3  
-r SampleECFCFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using only AS,X atomic invariants atom types in vector string format and create a SampleECAIFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a AtomicInvariantsAtomTypes  
--AtomicInvariantsToUse "AS,X" -r SampleECAIFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using only HBD,HBA functional class atom types in vector string format and create a SampleECFCFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a FunctionalClassAtomTypes  
--FunctionalClassesToUse "HBD,HBA" -r SampleECFCFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using atomic invariants atom types in vector string format and create a SampleECAIFP.csv file containing compound ID from molecule name line along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a AtomicInvariantsAtomTypes  
--DataFieldsMode CompoundID -CompoundIDMode MolName  
-r SampleECAIFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using functional class atom types in vector string format and create a SampleECFCFP.csv file containing compound IDs using specified data field along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a FunctionalClassAtomTypes  
--DataFieldsMode CompoundID -CompoundIDMode DataField --CompoundID Mol_ID  
-r SampleECFCFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using atomic invariants atom types in vector string format and create a SampleECAIFP.tsv file containing compound ID using combination of molecule name line and an explicit compound prefix along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a AtomicInvariantsAtomTypes  
--DataFieldsMode CompoundID -CompoundIDMode MolnameOrLabelPrefix  
--CompoundID Cmpd --CompoundIDLabel MolID -r SampleECAIFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using functional class atom types in vector string format and create a SampleECFCFP.csv file containing specific data fields columns along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a FunctionalClassAtomTypes  
--DataFieldsMode Specify --DataFields Mol_ID -r SampleECFCFP  
-o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using atomic

invariants atom types in vector string format and create a SampleECAIFP.tsv file containing common data fields columns along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a AtomicInvariantsAtomTypes
--DataFieldsMode Common -r SampleECAIFP -o Sample.sdf
```

To generate extended connectivity fingerprints corresponding to neighborhood radius up to 2 using functional class atom types in vector string format and create SampleECFCFP.sdf, SampleECFCFP.fpf and SampleECFCFP.csv files containing all data fields columns in CSV file along with fingerprints vector strings data, type:

```
% ExtendedConnectivityFingerprints.pl -a FunctionalClassAtomTypes
--DataFieldsMode All --output all -r SampleECFCFP
-o Sample.sdf
```

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SEE ALSO

InfoFingerprintsFiles.pl, SimilarityMatricesFingerprints.pl, AtomNeighborhoodsFingerprints.pl, MACCSKeysFingerprints.pl, PathLengthFingerprints.pl, TopologicalAtomPairsFingerprints.pl, TopologicalAtomTorsionsFingerprints.pl, TopologicalPharmacophoreAtomPairsFingerprints.pl, TopologicalPharmacophoreAtomTripletsFingerprints.pl

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